PROBLEM SET 7, PART 1: DE RHAM COHOMOLOGY GROUPS
DUE: DEC. 28

(1) [**An infinite dimensional H^1_{dR}**]
   (a) Find a closed 1-form on $\mathbb{R}^2 - \{(0,0)\}$ that is not exact.
   (b) Let $M = \mathbb{R}^2 - \mathbb{Z}^2$. Prove: $H^1_{dR}(M)$ is infinitely dimensional.

(2) [**The de Rham cohomologies of the union**]
   (a) Let $M = M_1 \cup M_2$ be the disjoint union of two smooth manifolds. Find the relation between $H^k_{dR}(M)$ and $H^k_{dR}(M_i)$.
   (b) Calculate the de Rham cohomology groups of S^1 via Mayer-Vietoris sequence.
   (c) What if $M = \cup_{i=1}^{\infty} M_i$ is the disjoint union of countably many smooth manifolds?

(3) [**Witten deformation**]
Let M be a smooth manifold, and $\theta \in \Omega^1(M)$ be an exact 1-form.
   (a) For any k, define $d_\theta : \Omega^k(M) \to \Omega^{k+1}(M)$ by

$$d_\theta(\omega) = d\omega + \theta \wedge \omega.$$

   Prove: $d_\theta(d_\theta \omega) = 0$ for all $\omega \in \Omega^k(M)$.
   (b) Consider the complex

$$0 \to \Omega^0(M) \xrightarrow{d_\theta} \Omega^1(M) \xrightarrow{d_\theta} \cdots \xrightarrow{d_\theta} \Omega^{m-1}(M) \xrightarrow{d_\theta} \Omega^m(M) \to 0.$$

   Define $Z^k_\theta(M)$, $B^k_\theta(M)$ and $H^k_\theta(M)$, and prove: $H^k_\theta(M)$ is isomorphic to $H^k_{dR}(M)$.

(4) [**Missing parts in the proof of Mayer-Vietoris sequence**]
   (a) Prove Proposition 6.2.1.
   (b) (NOT required) To prove Theorem 6.2.4, we need to prove six inclusion relations. We proved one in class. Try to prove the rest five relations.

(5) [**Applications of Mayer-Vietoris sequence**]
   (a) Let M_1, M_2 be connected manifolds, and $M = M_1 \# M_2$ be their connected sum. Find the relation between $H^k_{dR}(M)$ and $H^k_{dR}(M_1), H^k_{dR}(M_2)$.
   (b) Compute the de Rham cohomology groups of $\mathbb{RP}^2$ and $\mathbb{CP}^2$.
   (c) (NOT required) Compute the de Rham cohomology groups of $\mathbb{RP}^n$ and $\mathbb{CP}^n$.

(6) [**Kunneth formula and its applications**]
   (a) Read §6.2.2 "Application 3: Kunneth formula".
   (b) Prove $H^k_{dR}(\mathbb{T}^n) \simeq \mathbb{R}^{\binom{n}{k}}$.

(c) For any compact manifold M, define $p_M(t)$ to be the polynomial $p_M(t) = \sum_{i=0}^{n} b_i(M)t^i$. Prove: If M, N are compact, then

$$p_{M\times N}(t) = p_M(t)p_N(t).$$

(In particular, we get $\chi(M \times N) = \chi(M)\chi(N)$.)

(d) (NOT required) Prove: $S^m \times S^n$ is homeomorphic to $S^{m'} \times S^{n'}$ if and only if $\{m, n\} = \{m', n'\}$.

(7) **[Not required]** Let G be a compact connected Lie group acting smoothly on a smooth manifold M (from left).

(a) For each k, define left-invariant k-forms on M, and then define the "k^{th} left-invariant de Rham cohomology group" $H^k_L(M)$.

(b) Prove $H^k_L(M) \simeq H^k_{dR}(M)$, as follow:

(i) Let $i : \Omega^k_L(M) \hookrightarrow \Omega^k_{dR}(M)$ be the inclusion map . Prove: i induces a linear map $i_* : H^k_L(M) \to H^k_{dR}(M)$.

(ii) Prove i_* is injective.

Hint: Let dg be a normalized Haar measure on G, in other words, dg is the measure associated to a volume form α on G which is both left and right invariant, such that $\int_G \alpha = 1$. [You may try to prove the existence of such dg if you want.] For each $\omega \in \Omega^k(M)$ define the averaging of ω with respect to G to be

$$A(\omega) = \int_G \tau_g^*\omega \; dg.$$

Show that A is a linear map from $\Omega^k(M)$ to $\Omega^k_L(M)$, which induces a linear map $A_* : H^k_{dR}(M) \to H^k_L(M)$. Moreover, prove $A_* \circ i_* = \mathrm{Id}$.

(iii) Prove: i_* is surjective.

Hint: It's enough to prove $[A(\omega)] = [\omega]$ for any $[\omega] \in H^k_{dR}(M)$. First notice that the map A above can be rewritten as

$$A(\omega) = \int_G \tau^*\omega \wedge \pi^*\alpha,$$

where $\tau : G \times M \to M$ is the action of G on M, and $\pi : G \times M \to G$ is the projection, and one regards the differential form $\tau^*\omega \wedge \pi^*\alpha$ as a top form on G (with M variables as parameters). Take a contractible neighborhood U of e in G, and a top form β on G which is supported in U and satisfies $\int_G \beta = 1$. Then there exists a differential form η on G so that $\alpha - \beta = d\gamma$. (Why?) Let $\tau_U : U \times M \to M$ be the restriction of τ. Then $\tau^*\omega \wedge \pi^*\beta = \tau_U^*\omega \wedge \pi^*\beta$. Finally prove $\tau_U^*\omega = \pi_M^*\omega + d\eta$ for some differential form η on $U \times M$, where $\pi_M : U \times M \to M$ is the projection.

(c) Prove: $H^k_{dR}(\mathbb{T}^n) \simeq \mathrm{span}\{dx^{i_1} \wedge \cdots \wedge dx^{i_k} \mid 1 \le i_1 < \cdots < i_k \le n.\}$.